INNOVATION PROGRAMME 2026

Explore the **World of Design!**



Students' Pack

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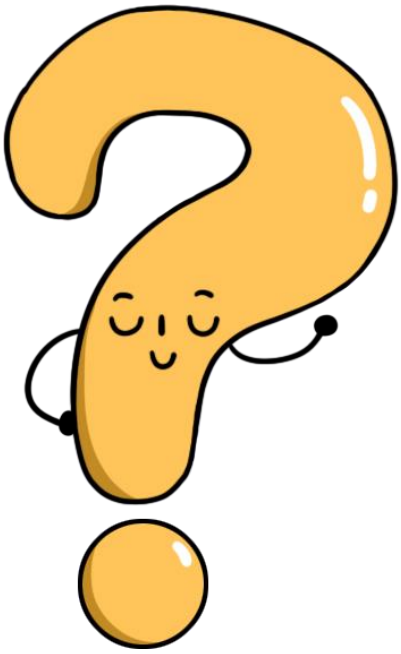
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MOE Innovation Programme

Part 1: About IvP 2026

Foreword



To understand what innovation is about, let us first begin with a true story that took place way back in 1972. One evening, Becky Schroeder, a ten-year-old girl, was doing her homework in the car while waiting for her mother to finish her shopping. As it grew darker, Becky had trouble seeing. She did not have a flashlight, and did not want to open the door to light up the whole car. So she thought it would be neat to have the paper light up somehow. That night, Becky went home trying to imagine different ways of making paper glow in the dark. First, she tried to use fluorescence, but she found that fluorescent things glow only when light falls on them so they would not have worked in the circumstance she was in. Next she thought of fireflies that glow due to bioluminescence in their bodies but this did not seem to be something she could incorporate easily into a piece of paper. Then she recalled the glow-in-the-dark Frisbees in her toy box which used phosphorescence. The next day she went to the hardware store with her father and bought some phosphorescent paint. That night she worked in her darkened bathroom painting stacks of paper, even screaming at one point, “It works! It works!” But she soon found that it wasn't the paper that needed to glow, it was the clip board. She coated her clip board with the paint, so that it gave off just enough light to shine through the paper clipped on top of it.

Over the next couple of years, from age ten to twelve, Becky refined her device to use batteries to sustain the glow. In 1974, her simple invention, which she named ‘The Glo-Sheet’, enabled her to become the youngest female ever to receive a U.S. patent. Her product sold briskly. Photographers used them in their darkrooms, critics found them perfect for taking notes in dark theatres, and emergency medical technicians used them in ambulances. Even NASA came knocking at her door. They had been developing something similar, and noted that if she were a former employee the patent would belong to them.

In the story, Becky Schroeder demonstrated the spirit of tenacity as she worked tirelessly to solve a problem which she experienced. She thought about the problem critically and drilled down to the crux of it. She did not let the limits of know-how at that time to stand in her way, but instead explored possibilities from nature, and saw the solution from a different perspective – that it was the clipboard that needed to

glow, not the paper. Indeed, these processes of thought and action in problem solving, and drive to see to the fruition of a solution is what the Innovation Programme hopes to nurture in you, our young innovators.

We wish all of you an exciting and fulfilling experience in the IvP this year.

The Innovation Programme Committee
Gifted Education Branch, CPDD, MOE



About the 2026 IvP

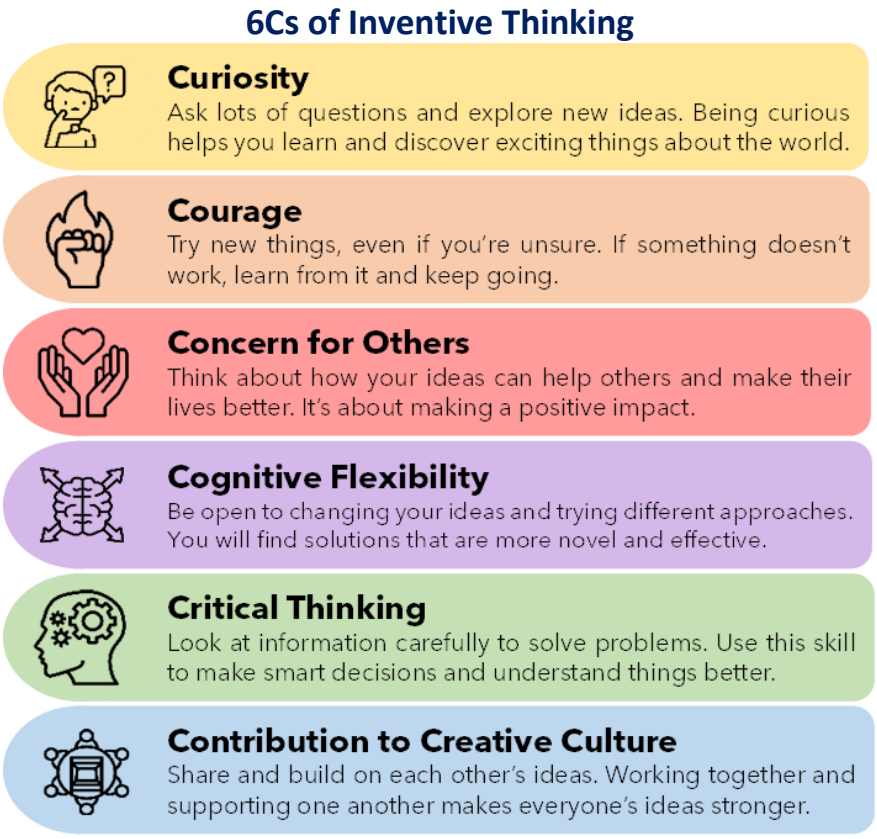
Join the Adventure of Innovation!

Since its beginnings in 1993, the Innovation Programme (IvP) has been an incredible journey for students like you to explore new possibilities, develop inventive thinking skills, and dive into experiences beyond the classroom. Whether you're in Primary 5 or Secondary 2/3, the IvP connects you with expert innovation mentors from universities and organisations who can help you turn your ideas into impactful solutions.

We believe that **you have the power to be a creator, a connector, and a contributor.** Through IvP, you'll grow in confidence, share your ideas, and learn how to give and receive feedback that sparks amazing outcomes—not just for yourself but for your entire team.

Discover the 6 Cs of Inventive Thinking!

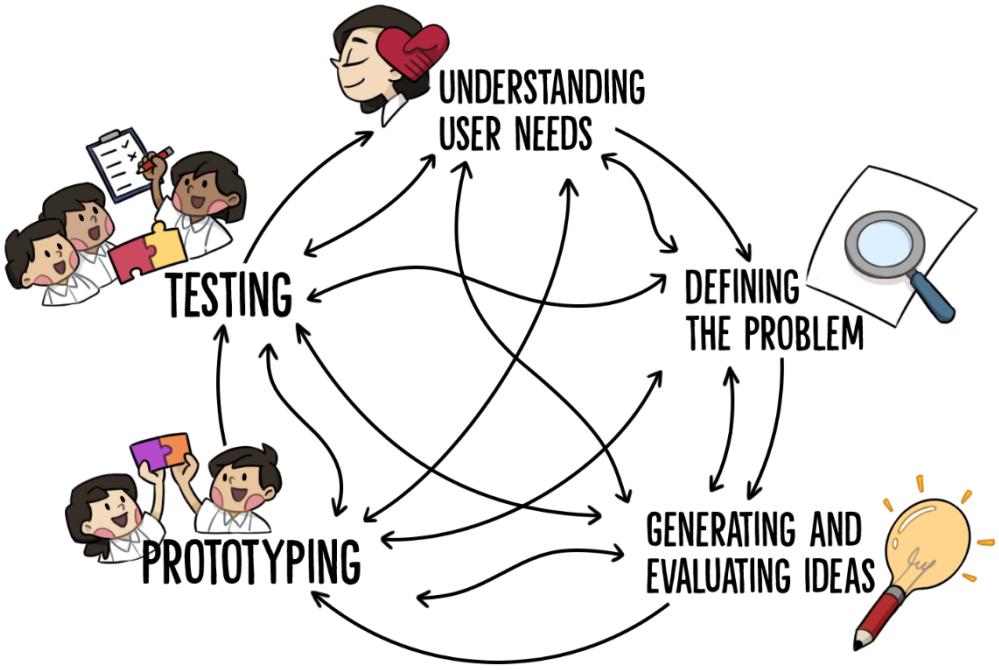
At the heart of IvP are six key skills and mindsets—the *6 Cs of Inventive Thinking*. These are your tools for understanding problems, generating fresh ideas, and designing solutions that make a real difference.



Your Innovation Toolkit

To help you master these 6Cs, we've created exciting lesson resources and hands-on innovation challenges! These activities are designed to spark your curiosity, unleash your creativity, and guide you step-by-step through the **Design Thinking (DT)** process. Ready to make your mark? Discover how your ideas can inspire change and build a brighter future!

Design Thinking Process



IvP Timeline

Your IvP Journey: From Start to Shine!



The **IvP cycle** begins with your school-based launch and reaches its exciting finale at the **Young Innovators' Fair.** Along the way, you'll receive regular IvP lessons and personalised mentoring from your teachers. But that's not all—Innovation Mentors from leading institutions will meet with you to share expert advice and unique insights to elevate your project ideas.

Here's what you'll do throughout your IvP journey:

- **Prepare for success:** Stay organised by filing your work and documenting your progress.
- **Learn from the best:** Dive into sessions with mentors and teachers who will guide you step-by-step.
- **Showcase your growth:** Create a report that captures your incredible learning journey and innovative solutions!



Part 2: About Innovation Challenges 2026

Project Requirements

This year, you have the exciting opportunity to choose between two innovation challenges, focusing on the area that sparks your interest most. Whatever you choose, your journey is just as important as your final project!

We encourage you to find creative ways to present your final project report—go beyond the usual and make it a story worth sharing. Show us your design thinking process in action, from research to your innovative solution!

Project Submission Guidelines

Teachers will guide you in crafting your submissions to effectively showcase your journey and final solution. Be sure to include these key components:

Category	Things to consider
Problem Identification and Understanding	• Demonstrate how you identified and understood the user's needs. • Include evidence from your research and observations that shaped your problem definition.
Ideation and Development	• Share the brainstorming journey: How did you generate ideas and progress towards a user-centred solution? • Let your creativity and thought processes shine here!
Prototyping and Testing	• Show how you brought your ideas to life through prototypes. • Include records of user testing and how feedback helped refine your solution.
Final Solution	• Present your finished product in a clear and engaging way. • Highlight its innovative features and how it addresses user needs.
Reflection	• Share insights from your learning journey: the challenges faced, the solutions you explored, and how your project grew from start to finish.

Innovation Challenge #1: User-first or User-fail?

This year, there are two Innovation Challenges: "User-first or User-fail?" and "Eureka or Err?' Each challenge offers a unique focus and approach. The first challenge, **"User-first or User-fail?"**, is all about keeping the user at the heart of everything you create. You'll explore how prioritising the user's needs throughout the design process can make the difference between a solution that works and one that doesn't.



- **User-first:** This means designing with empathy and understanding, ensuring your solution is practical, functional, and truly useful.
- **User-fail:** This happens when a design overlooks or misinterprets user needs, resulting in a product that might not work well or be adopted.

By taking on this challenge, you'll learn to design with empathy, keeping the user's perspective front and centre. You'll not only address real-world problems but also find the balance between creativity and practicality, ensuring your designs are impactful and resonate with their intended users. This challenge isn't just about solving problems—it's about creating meaningful and usable solutions that make a real difference. Are you ready to step into the shoes of your users and create something incredible?

For this challenge, we will be exploring these questions:

- Who is the user, and what problem are we solving for them?
- How can we be sure our design is solving the right problem for the user?
- How can we make our design useful, easy to use, and something the user will like?

Some Ideas for Exploration

Dive into real-world challenges by making users the stars of your designs! Imagine the people around you—what do they need? What problems do they face daily? Here are some inspiring ideas to kickstart your journey into user-centred innovation:

User: Someone Like You



1. **Interactive Learning Spaces Challenge:** Think about your ideal study area. How can you redesign or create a space that helps you focus and collaborate in creative ways? Consider tools, designs, or even furniture that could make learning spaces more dynamic and tailored to your needs.

2. **Sports Engagement Challenge:** What makes sports exciting? Create a new, fun concept for sports practice that gets everyone involved! How can you develop a game or tool that keeps participants motivated and energized, no matter what skill level they are at?

3. **Active Recess Challenge:** Recess is more than just a break! Design a completely new game or tool that transforms break time into a fun, active adventure. What activity could bring your friends together, help everyone stay fit, and be the highlight of your school day?

4. **Collaborative Learning Challenge:** Group work is a mix of fun and challenge. How can you design a tool or strategy that makes group work easier and more exciting for everyone? Think of ways to break down barriers, improve communication, and make participation a breeze.

5. **Focus and Stress Relief Challenge:** Imagine a tool that helps you stay calm and focused during study time. Design something simple, practical, or creative that would help your classmates stay relaxed and boost their concentration. What would help you or your peers reduce stress while studying?

User: Someone You Care For



1. **Helping a Family Member Challenge:** Everyone has chores or tasks that they struggle with. What if you could design something that made it easier for a family member (like a sibling or parent) to get things done? How could your solution take the stress out of everyday tasks and make things more manageable?

2. **Supporting a Friend Challenge:** Friends face challenges too. How could your design help a friend navigate tricky social situations, like making new friends or handling bullying? Think about tools, gadgets, or strategies that could improve communication and bring people closer together.

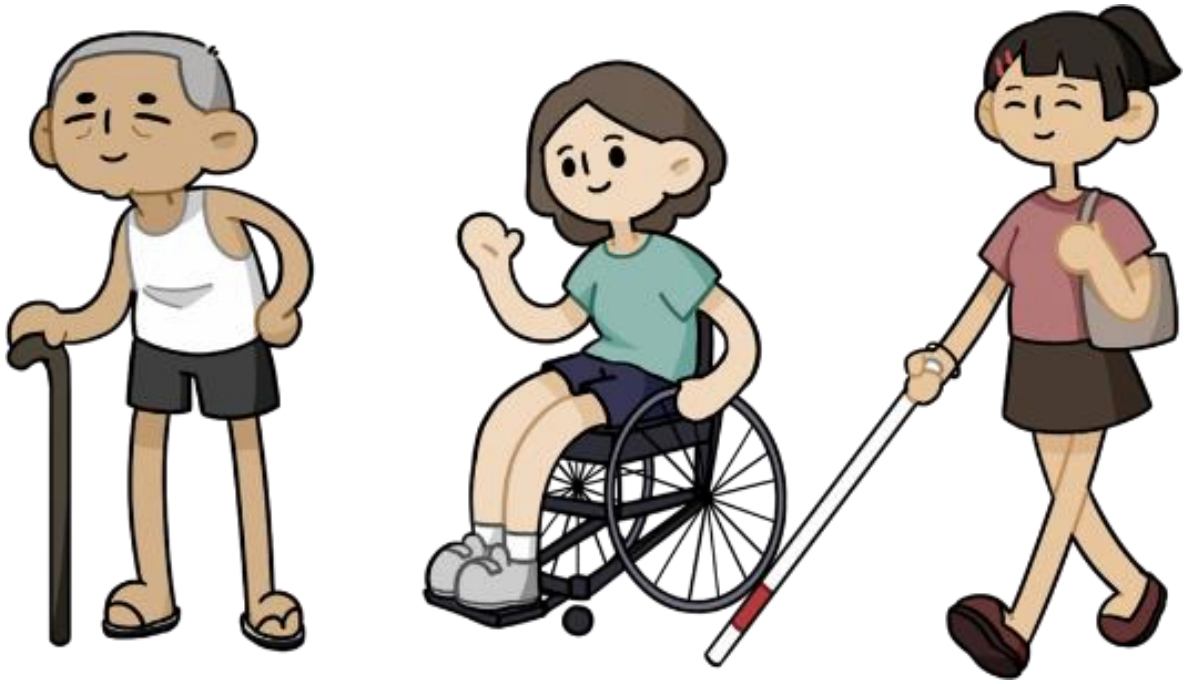
3. **Caring for a Sibling Challenge:** Siblings often need a little help, whether it's with daily tasks or navigating tough times. What could you design to make your sibling's life a little easier and less stressful? Consider practical solutions that could take the load off and help them handle tasks they find overwhelming.

4. **Improving Neighbourhood Safety Challenge:** Communities can always be safer and more welcoming. What clever design can you come up with to

make your neighbourhood safer? Whether it's a new technology or a fresh initiative, think about how you can help everyone feel more secure and confident in their surroundings.

5. **Helping a Teacher Challenge:** Teachers face a lot of challenges too! How can you design a tool that would help a teacher manage their class better? Focus on things that would reduce stress, improve organisation, or increase productivity, all while making the teacher's job a little more enjoyable.

User: Someone Different



1. **Helping People with Disabilities Challenge:** How can you make daily life easier for someone with disabilities? Design an invention or tool that would help them with something simple—like reaching items on a shelf, or walking to the store. Be creative! How can your design make their world more accessible and independent?

2. **Connecting the Elderly Challenge:** What happens when elderly people can't connect with others? How can you design a tool or system that helps seniors stay in touch with their community and feel included? Think about things that could help them chat with friends, get out of the house, and combat loneliness.

3. **Assisting People in Poverty Challenge:** What if your design could make a real difference in someone's life? How can you create a tool or system that helps people living in poverty improve their living conditions? Maybe it's an

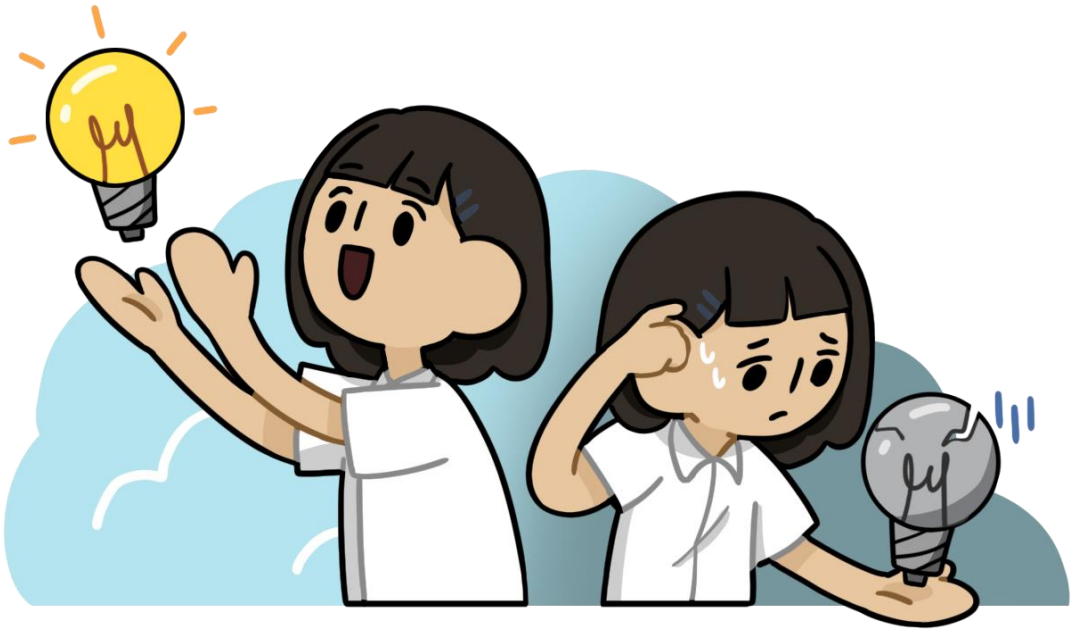
affordable solution to give them access to daily necessities, or something to help them find new opportunities for growth.

4. **Bridging Cultures Challenge:** In our diverse world, cultural connections are important! How could you design a tool that helps people from different cultures better understand and relate to each other? Think about solutions that could bridge gaps in communication, understanding, or shared experiences.

5. **Improving Mobility Challenge:** Mobility challenges affect millions of people. What innovative solution can you create to help those with physical limitations move around with greater ease? It could be anything from a new walking aid to an idea that enhances how they get around in public spaces. Be bold and think of something that makes a real difference.

Innovation Challenge #2: Eureka or Err?

The second challenge, **"Eureka or Err?"**, is all about embracing the creativity and iteration that drive the design process. It's an opportunity to push the boundaries of innovation by daring to dream big, learning from mistakes, and constantly improving your ideas.



- **Eureka Moments:** These are those flashes of inspiration where groundbreaking ideas take shape—solutions that are original, effective, and perfectly tuned to meet real needs.
- **Err Opportunities:** Mistakes and setbacks during the journey can be just as valuable. They highlight areas for improvement and guide you to refine your design, making it even stronger.

By taking on this challenge, you'll discover that innovation is not a straight line—it's a process that blends moments of triumph and lessons from failure. By balancing your creativity with a problem-solving mindset, you'll be empowered to deliver meaningful, user-focused solutions that truly make a difference. Get ready to celebrate the power of "Eureka" while learning from every "Err"!

For this challenge, we will be exploring these questions:

- Why is thinking in new ways important for design?
- How do we learn and improve from both successful and flawed ideas?
- What makes a design innovative and effective?

Some Ideas for Exploration

Ready to tackle everyday challenges with fresh, creative solutions? Here's your chance to think BIG, explore BOLD ideas, and transform ordinary problems into extraordinary innovations. Use your imagination, try out new approaches, and keep improving your designs to make them practical, impactful, and fun. Let's get started!

Place: Home



1. **Space-Saving Furniture:** Transform small spaces with furniture that adapts to your needs! Imagine modular seating that converts into a work desk or a sleek coffee table with hidden compartments to store your essentials in style.

2. **Sustainable Cooking Tools:** Reinvent the tools in your kitchen with eco-friendly innovation! Think of biodegradable scrubbing pads, compact appliances that cater to smaller servings, or energy-efficient timers to cut waste and promote sustainable living.

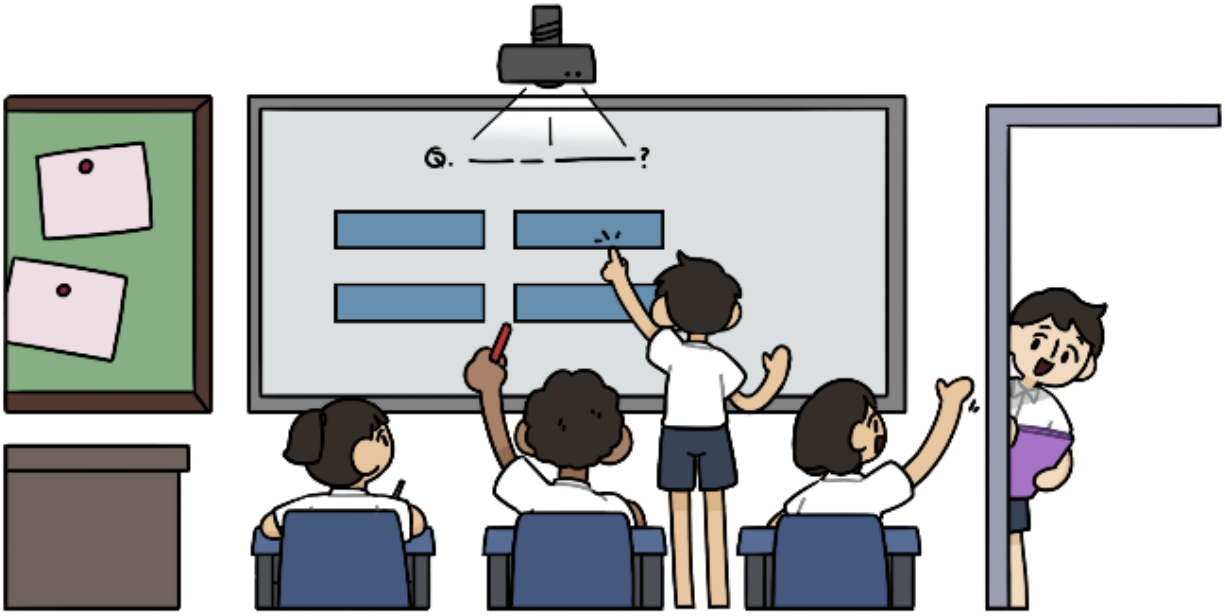
3. **Clean-Up Solutions:** Simplify your cleaning routine with tools designed for efficiency. Imagine a broom that cleans itself, reusable microfiber surface cleaners, or gadgets that eliminate the need for disposable wipes.

4. **Energy-Efficient Appliances:** Design smarter household gadgets that consume less energy. From motion-activated lighting to timers that optimize

appliance use, focus on creating compact, energy-efficient solutions to make homes greener.

5. **Home Organisation:** Take tidying up to the next level with innovative storage and organization solutions. Picture modular shelving with built-in charging docks or an automated system that sorts and folds laundry to save time and effort.

Place: School



1. **Fun Learning Tools:** Reimagine classroom activities with tools that educate while entertaining. Develop interactive games, hands-on projects, or apps with rewards and challenges to make learning irresistibly fun.

2. **Study Spaces:** Go beyond the standard desk setup by creating dynamic study environments. Design portable stations, modular seating, or calm corners tailored to different learning styles to keep students inspired.

3. **Classroom Set-Up:** Redesign the classroom into a hub of creativity and collaboration! Explore layouts with movable seating for group discussions, designated tech zones, or specialized spaces for creative exploration.

4. **Eco-Friendly Supplies**: Give school supplies a green makeover by redesigning them for sustainability. Envision longer-lasting materials, multi-use notebooks, or classroom systems that cut down on daily waste.

5. **Break Times:** Reimagine downtime with engaging break activities. Design outdoor play tools, digital relaxation guides, or team-based challenges that energize students and encourage social interaction.

Place: Neighbourhood



1. **Waste Solutions:** Make recycling accessible and exciting with innovative ideas. Imagine user-friendly sorting stations, neighbourhood drop-offs with reward systems, or creative campaigns to inspire waste reduction.

2. **Easy Gardens:** Transform barren spaces into lush, thriving gardens with low-maintenance solutions. Consider drought-resistant plants, vertical gardening kits, or water-efficient irrigation systems to suit busy lifestyles.

3. **Community Help:** Foster neighbourly collaboration with tools that bring people together. Develop apps for sharing tasks, lending tools, or coordinating local projects to strengthen bonds within the community.

4. **Fun Events:** Bring neighbourhoods to life with vibrant events! From outdoor movie nights to art festivals or sports tournaments, focus on inclusive designs that encourage everyone to join the fun.

5. **Safe Spaces:** Create safer, more inclusive communities with thoughtful designs. Imagine better-lit streets, community gathering areas, or programs that promote mentorship and camaraderie to make everyone feel secure and welcome.

Assessment

School-based Assessment & Descriptors



Innovation isn't just about the final product—it's about the journey! As we guide you through your creative process, we'll celebrate your growth, your courage, and your newfound skills. Ask yourself:

- **Do I feel more confident about innovating now?**
- **Have I faced my fears of judgment or failure, and taken bold steps to try something new?**
- **Have I built skills and mindsets that I can apply to future challenges?**

To answer these questions, we assess both the process and product domains:

- **Process Domain:** This focuses on your journey—how you approach problem-solving, the depth of your user understanding, and the resilience you show during iteration and testing.
- **Product Domain:** This focuses on the outcome—your solution's impact, functionality, and alignment with users' needs.

Descriptors Unpacked

To help you understand what we're looking for, here are the descriptors unpacked:

	Descriptors	Unpacked
Process	Consistently engage in research to understand ideas and enhance them significantly.	Research helps students understand users, their needs, and the problem more clearly. Consistent research means always gathering information to refine ideas. It includes looking for more information to understand the problem, interviewing, observing, and studying existing solutions.

	Actively try new ideas and learn from failure to make significant improvements.	When students actively try new ideas, they are experimenting with different approaches to solve a problem or create something new. It involves brainstorming, suspending judgment, and trying to make ideas work.
	Actively seek and apply feedback to make ideas more effective.	Actively seeking and applying feedback involves testing ideas with peers, teachers, or experts to gain different perspectives and insights into the designs and making changes based on the feedback to achieve the desired outcomes.
Product	Is highly effective and beneficial for its users. (Note: It could also positively impact humanity.)	The design addresses the identified problem or need, is easy and intuitive for users to use, and improves their overall experience. Additionally, the design can positively impact humanity at large, beyond individual user benefits.
	Is simple, aesthetic and provides significant improvements over existing ideas.	Simplicity ensures easy understanding and implementation, while aesthetics enhance visual appeal. The solution must significantly improve on current ideas, offering better functionality, efficiency, or user satisfaction.
	Is novel and resonates well with what people like and value.	A novel solution can capture attention and stand out, but it must also connect with what people find appealing or important.

Using the Assessment Rubrics

The assessment rubrics are your roadmap to success in the Innovation Programme (IvP). They help you understand what is expected at each stage of the process and provide a clear path for improving your project. Here's how you can use the rubrics to guide your work and achieve your best:

Understanding the Rubrics:

The rubrics are divided into two key components: Product and Process. Each one plays a crucial role in determining your final score, so let's break it down:

- **Process:** This looks at how you approach the project, how you solve problems, and how you apply feedback along the way. The process isn't just about making the right decisions—it's about showing how you engage with the work, refine your ideas, and learn from setbacks. Documenting your process helps teachers see your growth as an innovator and provides you with valuable feedback to improve your work.

- **Product:** This represents the final solution that you come up with at the end of the project. The rubrics will assess the effectiveness, innovation, and user-centeredness of your design. To score well here, focus on ensuring that your solution addresses the user's needs, is easy to use, and offers meaningful improvements over existing solutions.

How the Rubrics Influence Your Grade:

Your final grade is based on the sum of both the process and product scores. To achieve a **Distinction** (6), you need to excel in both areas. Here's how the breakdown works:

- **Distinction** (6 points): Achieved by scoring highly in both the product and the process domains. This means your solution must be impactful, user-centred, and innovative, while your process should showcase continuous improvement, resilience in testing, and learning from feedback.

- **Merit** (5 points): A great outcome that still meets the core criteria but might not demonstrate the same level of excellence as a Distinction. Your design may be strong, but perhaps you didn't document your process as effectively or missed out on exploring certain areas of the problem.

- **Participation** (4 points): This indicates that you have made progress but might need to work on some areas. Your solution may not fully address user needs, or your process documentation may lack depth.

How You Can Use the Rubrics to Your Advantage:

- **Refer to the Rubrics Throughout the Project**: Don't wait until the end to check the rubrics. Use them as a guide at each stage of your project. For example, when you research your user, ensure that it's documented and deep enough to help shape your final solution. Keep track of how you iterate your design and be mindful of where you can apply more research, feedback, or improvements.

- **Focus on the Process**: Teachers will observe and track how you engage in learning—how you solve problems, experiment with ideas, and take feedback seriously. Remember to document every step. Take notes on your brainstorming sessions, research, design tests, feedback received, and iterations. The more evidence you provide, the clearer it will be how you've developed throughout the project.

- **Be Intentional with Your Product**: When designing your final solution, always refer back to the user's needs. Does your design solve a real-world problem in a way that's simple, functional, and innovative? If you're unsure, ask for feedback from your peers, teachers, or mentors and make sure to refine your product based on what you learn.

	Process Students…	**Product** The proposed solution…
Flourishing (3)	• Consistently engage in research to understand ideas and enhance them significantly. • Actively try new ideas and learn from failure to make significant improvements. • Actively seek and apply feedback to make ideas more effective.	• Is highly effective and beneficial for its users. (Note: It could also positively impact humanity.) • Is simple, aesthetic and provides significant improvements over existing ideas. • Is novel and resonates well with what people like and value.
Emergent (2)	• Conduct some research to understand and improve ideas, resulting in some enhancements. • Try new ideas occasionally and make noticeable but modest improvements. • Seek feedback occasionally and integrate it to some extent, leading to moderate improvements.	• Is starting to show potential effectiveness and benefits for its users, with room for improvement. • Is relatively simple and offers modest improvements over existing ideas. • Is somewhat novel, showing elements of originality, but may not resonate well with what people like and value.
Dormant (1)	• Rarely use research to understand or improve ideas, leading to minimal growth or enhancement. • Rarely try new ideas, resulting in minimal or no improvements. • Rarely seek feedback or struggle to apply it effectively, resulting in little or no enhancement of ideas.	• Is in its early stages and requires further development to demonstrate effectiveness and benefits for users. • Is somewhat complicated and does not offer improvements over existing ideas. • Lacks originality and innovation and does not resonate with what people like and value.

Project Submission Timeline

T3W9-T4W1: Evaluation Weeks

Final evaluations begin! Your work will be assessed in school using the innovation challenge rubrics. These rubrics show how your creativity and effort will be recognised. Review them carefully, aim high, and present your best ideas!

Next Steps for Top Projects!

T4W1 (Wednesday, 16 Sep): Nomination Day

If your project earns a **Distinction Award,** it could be nominated for IvP Project Judging at the **Young Innovators' Fair on 6 November!** Here's what this recognition brings:

- **Expert Feedback:** Present your ideas to industry leaders and discover how to push them even further.
- **Inspiration & Impact:** Have your work featured in the online **IvP Gallery,** encouraging other budding innovators like you!

Final Submissions: Make It Count!

T4W2 (Friday, 25 Sep): Submission Day

Schools will submit the Certificate Award Form and upload students' Final Project Reports to the assigned Google Drive folder. Your efforts will be celebrated with achievement certificates—Distinction, Merit, or Participation. Let your work shine, and leave a lasting mark on your IvP experience!

